

Requirement Specification Document

E-Commerce Order Management Database System

Week 1 – Requirement Analysis and Customer Database Module

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Project Title	Gym Management System
Week	Week 1

1. Introduction

1.1 Purpose

The purpose of this document is to analyze and specify the business requirements of the **Gym Management System**. The system is designed to manage gym members, trainers, memberships, workout schedules, attendance, and payments efficiently, ensuring smooth day-to-day gym operations..

1.2 Project Scope

The Gym Management System will enable gym administrators to manage member registrations, trainer details, membership plans, attendance records, workout schedules, and payment information. The system will be developed using a relational database with well-defined tables and relationships to ensure accurate data

1.3 Intended Users

- ☐ Gym Members – Register, renew memberships, and view workout schedules.
- ☐ Gym Trainers – Manage workout plans and monitor member progress.
- ☐ Gym Administrators – Manage members, trainers, memberships, attendance, and payments.

2. Business Requirements

e following are the key business needs the system must satisfy:

- Maintain accurate member information.
- Store trainer details and assigned members.

- Manage different membership plans.
- Track daily member attendance.
- Record membership payments and renewals.

3. Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement Description
FR-1	The system shall register all traffic sensors and cameras.
FR-2	The system shall collect real-time traffic data every few seconds.
FR-3	The system shall detect traffic congestion automatically.
FR-4	The system shall update traffic signal timing based on congestion.
FR-5	The system shall record accident information.
FR-6	The system shall notify emergency services immediately after accident detection.
FR-7	The system shall allow administrators to monitor live traffic status.
FR-8	The system shall generate daily and monthly traffic reports.

4. Non-Functional Requirements

- The system should update traffic information within **2 seconds**.

Security

- Only authorized users can access the system.

Reliability

- The system should operate continuously with minimal downtime.

Availability

- The application should be available **24×7**.

Scalability

- The system should support thousands of traffic sensors and vehicles.

Usability

- The dashboard should be simple and easy to use.

5. Customer Database Module – Data Requirements

Since this week's focus includes the Customer Database Module, the following entity and attributes have been identified:

Attribute	Data Type	Description	Attribute	Data Type	Description	Attribute	Data Type	Description
sensor_id	INT (Primary Key)	Unique sensor identifier	sensor_id	INT (Primary Key)	Unique sensor identifier	sensor_id	INT (Primary Key)	Unique sensor identifier
location	VARCHAR(100)	Sensor installation location	location	VARCHAR(100)	Sensor installation location	location	VARCHAR(100)	Sensor installation location
traffic_density	INT	Number of vehicles detected	traffic_density	INT	Number of vehicles detected	traffic_density	INT	Number of vehicles detected
signal_status	VARCHAR(20)	Green/Yellow/Red	signal_status	VARCHAR(20)	Green/Yellow/Red	signal_status	VARCHAR(20)	Green/Yellow/Red
vehicle_count	INT	Total vehicles counted	vehicle_count	INT	Total vehicles counted	vehicle_count	INT	Total vehicles counted
accident_status	VARCHAR(20)	Accident detected or not	accident_status	VARCHAR(20)	Accident detected or not	accident_status	VARCHAR(20)	Accident detected or not

6. Assumptions and Constraints

- ☐ The system uses a relational database (MySQL/PostgreSQL).
- ☐ Traffic sensors provide accurate real-time data.
- ☐ Internet connectivity is always available.
- ☐ Every traffic signal is connected to the central server.
- ☐ This document covers **Week 1** requirements only. Future modules such as Emergency Response, Vehicle Tracking, and Traffic Analytics will be developed in later weeks.

7. Conclusion

this document defines the initial business and functional requirements for the **Real-Time Smart City Traffic Management System**. The focus of Week 1 is the **Traffic Monitoring Module**, which serves as the foundation for developing the ER diagram, database schema, and future implementation.